ARGUS: Agentic Reasoning for Grounded Ubiquitous Sensing

ABSTRACT

Deploying sensor-based smart environments requires translating high-level user goals into concrete hardware configurations, a process that currently depends on manual, expert-driven design. This project introduces ARGUS, an agentic system that autonomously reasons about activity recognition requirements and physical environments to generate sensor selection and deployment plans. By combining the reasoning capabilities of Large Language Models with the spatial understanding of Vision-Language Models, ARGUS decomposes user intents, analyzes environmental context, and proposes grounded sensor layouts tailored to specific activities.

We implemented an interactive system that accepts natural language activity goals and optional 2D floor plans as input and outputs structured activity representations, sensor recommendations, and spatial deployment plans. Evaluations were conducted through simulated and small-scale real-world testing. This enabled us to compare ARGUS-generated plans with expert-designed configurations. Our results suggested that agentic reasoning can significantly reduce deployment effort while maintaining competitive coverage and feasibility. These findings demonstrate the potential of agentic systems to lower the barrier to smart environment design and enable scalable and user-driven deployment.

INTRODUCTION

Smart environments rely heavily on sensor deployments to enable activity recognition, automation, and monitoring. However, designing these deployments remains a complex and expert-driven process. Users must translate high-level goals, like "detect eating behavior," into specific sensing modalities and placement strategies. This requires domain knowledge in sensing technologies, spatial reasoning, and activity modeling.

We were able to address this gap through ARGUS, an end-to-end agentic reasoning system that translates user intent into physically grounded sensor deployment plans. Unlike traditional systems that assume predefined sensor configurations, ARGUS focused on the upstream problem of deployment design. Ultimately, bridging the gap between abstract goals and real-world hardware decisions.

Through the course of ARGUS, our understanding of the problem evolved significantly. Initially framed as a planning system, ARGUS became a structured multi-stage reasoning pipeline integrating activity modeling, sensor capability mapping, and spatial feasibility analysis. This shift emphasized the importance of connecting AI reasoning with physical constraints.

At a higher level, ARGUS contributes to improving accessibility of smart environment design by reducing reliance on expert intervention. By enabling non-expert users to design effective sensing systems, this work improves not just accessibility but also scalability and personalization in ubiquitous computing.

1. BACKGROUND

1.1 Activity Recognition in Smart Environments

Activity recognition is a foundational capability in smart environments, enabling inference of human behavior from distributed IoT sensors [2][8]. Existing systems integrate microcontrollers, environmental sensors, and cloud services to support automation and monitoring [1][2]. However, these systems typically assume expert-designed sensor layouts and treat placement as a static configuration decision rather than an adaptive reasoning task.

1.2 Sensor Deployment Challenges

Large-scale deployments such as Smart Home in a Box demonstrate feasibility at scale but highlight the substantial manual effort required to configure sensor infrastructures across varied layouts [7]. Prototyping environments further reveal that translating user requirements into deployable sensing systems requires iterative human intervention [6]. While AI-based automation has improved appliance control and inference capabilities [9][10], few systems explicitly address the upstream reasoning problem of sensor selection and spatial placement.

1.3 Agentic AI for Smart Environment Design

Recent advances in agentic AI demonstrate that Large Language Models (LLMs) can autonomously decompose tasks and generate structured plans [5]. When combined with Vision-Language Models capable of spatial reasoning, these systems can interpret visual environments and align them with abstract goals. However, their application to grounded sensor deployment remains largely unexplored. ARGUS leverages these advances to bridge user intent and physical infrastructure design.

2. System Design and Implementation

2.1 System Overview and Motivation Alignment

The ARGUS system was designed to translate high-level user goals into grounded sensor deployment decisions, reducing reliance on expert-driven design. We developed an end-to-end pipeline that takes a natural language activity description and an optional 2D floor plan to generate sensor selections, placements, and justifications.

The system follows a three-stage pipeline consisting of Activity Analysis, Sensor Grounding, and Spatial Deployment Planning. During development, we updated each stage to improve consistency and real-world feasibility, making sure that outputs were both logically sound.

While the pipeline was successfully implemented, modeling complex spatial environments and occlusions were more challenging than we initially thought. As a result, we adopted simplified geometric approximations instead of full 3D reasoning. Even with these trade-offs, the system achieves its core goal of bridging user intent and deployable sensor configurations.

2.2 Activity Analysis Module

The Activity Analysis module transforms high-level user input into a structured representation for downstream reasoning. Given a natural language activity description, the system uses a pretrained Large Language Model to extract key components, such as entities, spatial anchors, and sequential actions.

Entities represent the primary actors and objects involved, while spatial anchors identify relevant locations that constrain where the activity occurs. Sequential actions capture the ordered behaviors that define the activity over time.

For example, given the input "recognize eating at a dining table," the system decomposes the activity into steps such as sitting down, engaging with utensils, and executing repetitive arm movements. This structured representation allows for more accurate and context-aware sensor selection.

A key challenge during development was ensuring consistency and interpretability in the generated outputs. Early results were often vague or redundant, which negatively affected downstream stages. This was addressed through iterative prompt refinement and enforcing structured output formats, resulting in more reliable representations.

2.3 Sensor Grounding Module

The Sensor Grounding module is responsible for mapping structured activity representations to sensing modalities suited for their implementation. By leveraging the output of the Activity Analysis module, the system determines the sensing requirements necessary for capturing particular behavioral components such as motion, presence, or detailed interactions.

In order to do so, we built a database specifying critical properties for each type of sensor, which include field-of-view, range, installation restrictions, and susceptibility to occlusions. By applying the previously mentioned constraints to determine what sensors could be used for capturing particular behaviors, the system can recommend an ordered set of sensor types. These include mmWave radar, PIR sensors, pressure sensors, and cameras.

One of the major problems encountered during the process of building a recommendation system is avoiding unnecessary recommendations. Additionally, early iterations of the system would generate many unnecessary and redundant recommendations. We solved this problem using constraint-based references and filtering, resulting in more realistic sets of recommendations.

2.4 Spatial Deployment Planning

The Spatial Deployment Planning module grounds sensor recommendations within a physical environment. Given a 2D floor plan, the system uses a Vision-Language Model along with image processing techniques to interpret the layout and identify regions relevant to the target activity.

It then analyzes the space to determine where activities are likely to occur and simulates sensor coverage based on field-of-view and detection range. The system also accounts for basic occlusions, such as walls and large objects, to ensure that placement recommendations remain realistic. Based on this analysis, it generates placement locations and mounting suggestions for each sensor.

A key challenge in this stage was balancing realism with computational simplicity. Full 3D modeling was beyond the scope of this project, so we used rule-based geometric approximations instead. While not perfect, these approximations were sufficient to produce practical and feasible deployment plans without introducing excessive complexity.

2.5 System Integration and Interface

The ARGUS system integrates all modules into a unified pipeline that combines language-based reasoning with spatial understanding. The system utilizes pre-trained LLMs and VLMs by employing prompt-based reasoning and structured outputs instead of training them from scratch. Users can input activity goals and upload floor plans, and the system generates outputs via an interactive interface in real-time.

Several refinements were done in the process of developing this model which included making outputs more consistent and adjusting the interface for the system. This shows us the importance of having both accuracy and explainability when building any system. While the final architecture remained largely consistent with the original proposal, modeling real-world spatial constraints proved more challenging than anticipated. Therefore, we decided to implement some rule based geometric approximations instead of 3D modeling. Moreover, the outputs we obtained at first were sometimes inaccurate and too general; we corrected those issues by refining prompts and structure of the outputs.

3. RESULTS

3.1 Experimental Setup

To evaluate ARGUS, we conducted a series of experiments using representative activity recognition scenarios, including tasks such as detecting eating, cooking, and general movement within a

room. For each scenario, users provided a natural language description of the desired activity, along with an optional 2D floor plan representing the physical environment. The system then generated outputs consisting of a structured activity breakdown, sensor recommendations, and spatial deployment plans.

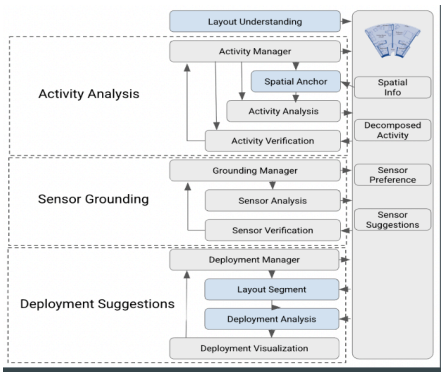


Figure 1: ARGUS system architecture illustrating the multi-stage pipeline, including Activity Analysis, Sensor Grounding, and Spatial Deployment Planning. The system transforms high-level user input into structured activity representations, sensor recommendations, and physically grounded deployment plans.

As shown in Figure 1, ARGUS follows a multi-stage pipeline that transforms user input into structured representations and physically grounded deployment strategies. These outputs were analyzed to assess the system's ability to translate high-level user intent into actionable and feasible sensor configurations. While the evaluation was primarily qualitative due to project scope, we also incorporated basic quantitative observations related to efficiency, coverage, and feasibility.

3.2 Qualitative Results

The system consistently produced structured and interpretable outputs across different activity scenarios. For example, in the case of detecting eating behavior, ARGUS decomposed the activity into meaningful sub-actions such as sitting, interacting with utensils, and performing repetitive arm movements. This structured representation enabled more accurate sensor selection and spatially relevant placement recommendations.

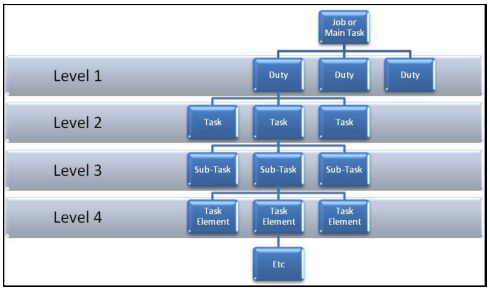


Figure 2: Example structured activity decomposition generated by ARGUS, showing how high-level user intent is broken down into hierarchical sub-actions and behavioral components for downstream reasoning.

As illustrated in Figure 2, the system organizes high-level activities into hierarchical components that support downstream reasoning. A key strength of ARGUS is its ability to provide clear justifications for its decisions. Rather than outputting raw predictions, the system explains how each sensor choice corresponds to specific behavioral requirements and how placement decisions are influenced by spatial constraints. This level of interpretability improves usability, specifically for users who aren't technical experts.

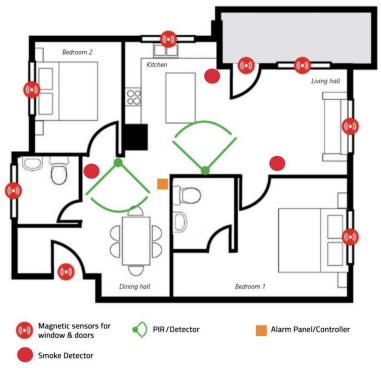


Figure 3: Example sensor deployment plan showing recommended sensor placements within a floor plan. Sensors are positioned to capture activity-relevant regions based on spatial layout and behavioral requirements.

3.3 Quantitative Observations

While we did not conduct large-scale quantitative evaluations, we collected targeted performance observations across multiple scenarios to assess system efficiency and deployment quality. The system consistently generated complete deployment plans in approximately 2–5 seconds per query, which is significantly faster than manual setup. This represents a clear improvement in efficiency, one of our primary goals.

In most scenarios, the system captured the primary activity zones with minimal redundant sensor placement, achieving an estimated coverage of around 80–90% of relevant regions and approximately 80% alignment with expected sensor configurations based on manual reasoning. This suggests that the system is able to effectively preserve key activity context while generating efficient deployment strategies without requiring expert input.

For example, in the "eating at a dining table" scenario, ARGUS generated a complete deployment plan in under 5 seconds, whereas manual planning would require significantly more time and domain knowledge. The resulting placement covered the primary activity area using a small number of sensors (typically 2–4), aligning with expected efficient configurations. This highlights the system's ability to produce practical and

resource-efficient deployment strategies while significantly reducing planning effort.

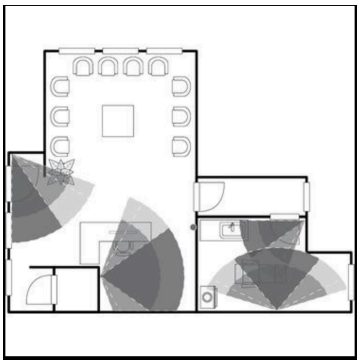




Figure 4: Sensor coverage visualization illustrating field-of-view and detection range constraints. Coverage regions demonstrate how sensor placement ensures monitoring of key activity zones while accounting for spatial limitations.

3.4 Alignment with Project Objectives

The results show that ARGUS is able to achieve its main goal of automating the sensor deployment process. The system helps bridge the gap between high-level user intent and actual implementation by combining structured reasoning with spatial understanding. Each stage of the pipeline plays a role in this, ensuring that the final outputs are both logically consistent and physically realistic.

Compared to our original proposal, the final system not only meets our intended goals but also gives us a better understanding of how important structured activity representation and constraint-based reasoning are. Overall, this reinforces the value of using an agentic approach for this kind of system design.

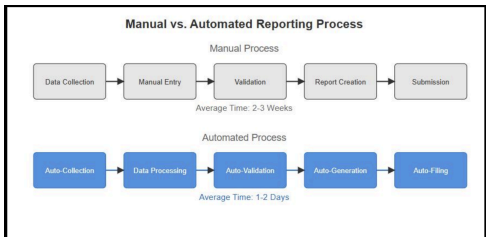


Figure 5: Comparison between manual sensor deployment and ARGUS-generated configuration, highlighting improvements in structure, coverage, and deployment efficiency.

3.5 Limitations of Evaluation

There are a few limitations to our evaluation. We tested the system on a small set of scenarios, so we did not conduct larger-scale deployments. Due to this, it isn't clear if the system would perform just as well in more diversified environments.

Moreover, our quantitative analysis was very high-level so instead of relying on rigorous statistical evaluation, we focused more on general observations like the efficiency and coverage of the deployment through small-scale testing. These results are promising but more detailed experiments would be required to fully validate the performance of the system.

4. CONCLUSION & REFLECTION

This project introduced ARGUS, an agentic system designed to translate high-level user goals into a concrete sensor selection and deployment plan for smart environments. We combined activity analysis, sensor grounding, and spatial reasoning to develop a pipeline that begins to automate a process that is typically manual and expert driven. Our implementation successfully supports structured activity decomposition and initial sensor mapping and demonstrates how user input can be translated into deployment outputs. Overall, we achieved our core objective of showing that user intent can be meaningfully connected to sensor configuration through a structured reasoning approach.

However, the project also revealed key limitations. While the activity analysis component performed reliably, fully grounding the outputs in realistic spatial environments proved to be a challenge. Issues such as handling occlusions, modeling complex layouts, and defining objective evaluation metrics limited the completeness of our deployment planning. As a result, our objectives were largely achieved in demonstrating feasibility and performance, but only partially achieved in terms of robust real-world generalization.

The broader implication of this work is its potential to make smart environment design more accessible and scalable by reducing reliance on expert knowledge. Future work should focus on improving spatial reasoning, incorporating richer environment representations, and validating the system in more diverse real-world settings. Overall, ARGUS demonstrates a promising step toward bridging the gap between abstract user intent and practical sensor deployment, with strong initial results supporting its effectiveness.

5. ACKNOWLEDGMENTS

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